

Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

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Abstract

Graph-based parameter transfer can give the quantum approximate optimization algorithm (QAOA) a useful starting point, while leaving improvements too small for finite-shot refinement to resolve. We study this interaction for MaxCut by separating reference selection, search geometry, model fitting, and statistical acceptance. A structural certificate bounds objective transfer, and local-model error bounds give sufficient acceptance margins. A finite-horizon analysis quantifies both valid acceptance and the suppression of small improvements by the prescribed confidence rule. In a 4,032-run ideal-circuit study on 24 test graphs, a disjoint 16-graph reference bank, and depths one and two, descriptor initialization meets the stated quality target on every graph. Nine undisplaced donor-based local-model policies consume 196 million optimization shots across 2,592 runs without an accepted step. A follow-up on 16 test graphs, two additional reference banks, and depths two and three includes explicit initialization-only controls. Empirical-Bernstein acceptance changes one isotropic and four shaped outputs out of 64 each; the matched isotropic Hoeffding rule changes none. SPSA and COBYLA attain lower mean reference deficits than every local-model policy at both follow-up depths. The demonstrated benefit is initialization. Achieving query-efficient refinement additionally requires proposals and measurement resolution matched to the available improvement; the evaluated pipeline does not establish a measurement-complexity advantage.

1 Introduction

The quantum approximate optimization algorithm (QAOA) searches a family of parameterized quantum states for a large expected objective value [11]. Its classical parameter dimension can be small at shallow depth, but each objective estimate requires repeated circuit measurements. Reusing parameters from a related graph offers an appealing way to reduce this cost. Once transfer produces a good starting point, however, the relevant question changes: can a finite measurement budget resolve the improvements that remain?

We examine this question for MaxCut through a graph-conditioned trust-region pipeline. Graph comparison selects a donor vector; nearby, symmetry-aligned donors set a bounded search shape; an affine model proposes a move; and fresh endpoint measurements determine acceptance. This construction makes four possible sources of benefit separately testable. An objective-estimation call, or query, is one newly sampled circuit estimate at one angle vector. We report its shot cost and its grouping into logical batches, because fewer calls need not imply fewer circuit executions or lower total cost.

Parameter transfer has several established foundations. Depth-dependent patterns motivate interpolation and Fourier initialization [40], and discretized annealing provides structured initial schedules [30]. Concentration results motivate fixed-depth reuse within specified sparse random

graph ensembles [2]. Local subgraphs, degree parity, and learned graph representations support graph-dependent donor selection [10, 12]. Offline local-tree calculations and precomputed angles can also reduce online parameter search [34, 38]. More recent approaches generate graph-conditioned parameter trajectories [25] or refine a selected layer after donor transfer [27]. These approaches differ in training, preprocessing, and online measurement costs; their common motivation makes initialization-only controls essential when assessing additional refinement.

Our structural comparator uses Gromov–Wasserstein (GW) distortion between graph shortest-path relations [24, 28], following the within-domain relational comparison used in SCOT [7]. A vertex coupling supplies a donor-selection score, while the donor angles supply parameter-space geometry. A separate rooted-edge-neighborhood comparison yields a uniform objective-transfer certificate. The distinction is substantive: the global shortest-path score is a heuristic, whereas the local certificate has explicit hypotheses and can be vacuous on particular graph pairs.

Local modeling and shot allocation also have substantial precedent. QR-based least squares and interpolation geometry control numerical sensitivity [5, 14]. Sung et al. use local quadratic models and observation reuse for variational quantum optimization [35]; ANASTAARS builds noise-aware models in adaptive random subspaces [8]. The iCANS optimizer allocates shots to gradient components [21], while the QAOA method of Ha, Shashaani, and Menickelly models both the objective and its spatial variance [15]. Stochastic trust-region and adaptive-sampling convergence theories impose their own accuracy and sampling conditions [4, 31]. Here the model is full-dimensional and affine, the graph-derived shape has fixed volume, and independent acceptance samples give a finite-budget decision guarantee. We compare controlled variants of this construction with SPSA and COBYLA; the related methods above provide context rather than additional measured baselines.

The contribution is a quantitative separation of transfer quality, proposal quality, and statistical resolution in one reproducible finite-shot study. The analysis connects interpolation error to a sufficient acceptance margin and bounds the probability that the prescribed Hoeffding policy detects a smaller margin. The experiments expose the corresponding distinction: descriptor transfer meets the canonical target before refinement, while nine donor-based refinement policies return the same anchors after substantial measurement expenditure. A follow-up with two additional banks and a variance-sensitive acceptance rule resolves a small number of moves, but SPSA and COBYLA retain lower mean deficits. These findings identify which part of the pipeline accounts for the observed quality and what remains to be demonstrated before claiming a query advantage.

2 Results

2.1 Separating initialization from refinement

The workflow in Fig. 1 exposes the comparison that a transfer claim needs. Returning the selected anchor is a complete initialization-only policy; fitting and accepting a new point incur additional target-graph measurements. The objective is normalized by the number of edges, with $f_G = -F_G/|E| \in [-1, 0]$, so a decrease in f_G is an increase in expected cut. Two quantities separate the two stages. The reported deficit $D_G = 100[f_G(\theta_{\text{out}}) - f_G^{\text{ref}}]$ measures the returned vector against the best-found reference, so smaller is better and negative values are possible because that reference is not a certified optimum. The incremental improvement $I_G = 100[f_G(\theta_0) - f_G(\theta_{\text{out}})]$ measures the returned vector against the policy’s own initialization, so $I_G > 0$ exactly when refinement improved on the anchor it started from and $I_G = 0$ whenever no step is accepted. Reporting both prevents a good anchor from being credited to the refinement algorithm.

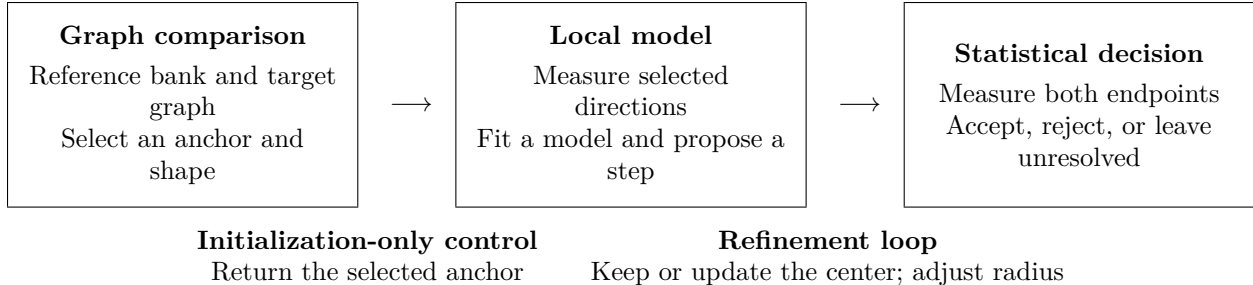


Figure 1: Graph information and measurement decisions enter at different stages. The reference bank is built offline. The local-model and decision stages consume the online shot budget. Independent final validation evaluates the returned vector for every policy, including initialization alone. A useful anchor does not by itself establish that refinement saves measurements.

2.2 Measured quality and resource use

The comparison separates initialization from refinement. The study comprises 4,032 optimization runs on 24 held-out graphs, using two depths, two shot caps, three shot seeds, and 14 methods. All graphs, settings, call records, donor fits, and final validation measurements are saved. The reference bank contains 16 nonisomorphic graphs; held-out graphs are also nonisomorphic to every bank graph. The same graph instances and resource caps are used across all policies in this ideal-circuit simulation study.

Initialization and region shape have distinct effects. At depth 2 and the larger shot cap, descriptor transfer with an isotropic region changes the mean normalized expected cut relative to the fixed-donor control by 1.05 (95% interval 0.65 to 1.51) percentage points, and relative to a random start by 13.95 (95% interval 12.23 to 15.78) percentage points. Keeping the descriptor initialization fixed, the shaped region changes it by 0.00 (95% interval 0.00 to 0.00) percentage points. GW with a shaped region changes it relative to descriptor transfer with an isotropic region by -0.52 (95% interval -0.84 to -0.23) percentage points. Positive values indicate higher expected cut for the method being compared against its stated baseline. Intervals resample the 24 graph-level means after averaging the three shot seeds; they are descriptive, unadjusted percentile bootstrap intervals, not simultaneous confidence statements. The full ablation table prevents an initialization benefit from being attributed to anisotropy (Table 1). The nine undisplaced donor-based local-model policies accept 0 refinement steps in 2,592 runs across both depths and budgets, despite consuming 196,341,504 online shots. For these policies, the final parameter vector equals the transferred anchor: their quality comes entirely from initialization. An explicit initialization-only comparison therefore gives the same depth-2 deficits, 0.38 percentage points for the descriptor anchor and 0.90 for the GW anchor, at zero online refinement shots. This comparison retains the offline donor-construction cost and does not waive any independent output-validation measurements. The 240 accepted moves occur only in the random-start, displaced-prior, and shuffled-reference controls. At the larger depth-2 budget, SPSA has a mean reference deficit of 0.11 percentage points, compared with 0.38 for descriptor-based local-model search and 0.90 for GW-based local-model search. The prescribed conservative refinement therefore has no demonstrated efficiency advantage over directly using the descriptor-selected anchor. The descriptor initialization already meets the prespecified target on 24/24 graphs at depth 1 and 24/24 graphs at depth 2, before target-graph measurements. The corresponding run counts are 72/72 and 72/72, because each graph contributes three identical initializations; repeated shot seeds add no independent graphs. Zero target-graph shots therefore

do not mean zero total cost: donor optimization is performed offline. The target is an expected normalized cut within 0.02 of an independently computed best-found depth-matched reference; it is not a certified QAOA optimum. A negative deficit means that a run exceeds that reference.

Conservative decisions consume the available information. Across all trust-region variants, 240 of 21,484 attempted decisions are accepted and 12,763 remain unresolved at their sampling cap or available budget. Unresolved decisions keep the current point. Figure 4 separates fitting from acceptance measurements. Fixed-shot decisions start at the largest planned sample count; adaptive decisions may stop at an earlier certified look. That single look spends 65,536 shots on the two endpoints, more than the smaller 32,768-shot cap, so the fixed-shot policy can take no acceptance measurement at that cap; 144 unresolved decisions are recorded with zero acceptance shots. Their confidence penalties differ because the fixed-shot variant has one planned look. These are finite-budget comparisons of complete policies, not a demonstration of asymptotic convergence.

An offline evaluation of the recorded endpoints finds 2,898 truly improving proposals among 16,299 trials of the eight adaptive undisplaced donor policies. Of those trials, 15,852 (97.3%) have true acceptance margin $h = f(x) - f(x^+) - \eta q \leq 0.005$; positive decreases are counted above a 10^{-12} numerical tolerance. For this margin range, Proposition 5 bounds the conditional probability of acceptance in any one trial by 3.48×10^{-6} , even if all four prescribed looks are affordable. The fixed-shot control is excluded from this calculation because it uses a different confidence penalty. These exact proposal values are retrospective diagnostics, never online inputs. The counts include dependent trials and are not independent observations or an empirical estimate of the acceptance probability; they connect the transferred-anchor experiment to the regime in which the proposition predicts low acceptance probability.

Building the reference bank used 14,560 exact objective evaluations and 2.3 seconds in the recorded environment. Those exact simulator calls have no assigned hardware-shot equivalent and must not be omitted from deployment considerations. Every final point receives 16,384 additional independent validation shots, one evaluation, and one simulated job. Tables show optimization cost separately so that the common validation cost can be added explicitly. Hardware-job counts describe the specified batching schedule; no device latency or hardware run is inferred. The structural transport comparisons require 9.13 seconds per target graph on average, shared between the two depth-specific banks. Descriptor selection and shape construction require 0.55 milliseconds at depth one. This substantial classical overhead provides an additional reason to prefer the simpler selector for this study; it is not converted into a hardware speedup. Of the 384 distinct target-reference transport solves, 253 end above the prescribed 10^{-9} coupling-change tolerance after the iteration cap. Their couplings satisfy the marginal constraints, but feasibility does not establish convergence or global optimality. These finite-iteration scores must be interpreted together with solver and donor-sensitivity diagnostics rather than as exact graph distances.

Model geometry and the graph bound have limited roles. For descriptor, shaped, the median local-design condition number is 3.99, with 95th percentile 3.99. For no point repair, the median local-design condition number is 17.23, with 95th percentile 223.85. At depth 1, 301/384 target-reference pairs have local-neighborhood total variation equal to one. At depth 2, 384/384 target-reference pairs have local-neighborhood total variation equal to one. At depth 2 those recorded scores are constant across the bank to within 10^{-15} , which is double-precision rounding in the edge-frequency sums and not structural information. The softmax floor $\tau = 10^{-3}$ in the selection rule nevertheless resolves that rounding, so the local-total-variation anchor differs from the fixed-donor anchor on 3 of 24 depth-2 targets, that is 9 of 72 runs per budget. The separation

Table 1: Complete depth-2 comparison at an optimization cap of 131,072 shots. Deficit is 100 times the signed difference from the best-found normalized reference objective. Each row averages 24 graphs and three shot seeds. Target success counts all 72 runs, including zero-shot initial successes. Capped target cost assigns the full budget to failures and uses the first successful recorded incumbent for successes. The first two rows are initialization-only policies that return the selected anchor and make no optimization call, so refinement gains are never credited to transfer. Shots exclude 16,384 independent final-validation shots per run, which every policy including initialization alone still incurs. The Local TV and Fixed donor rows differ only through a floating-point tie-break among depth-2 scores that are all equal to one; see the text.

Method	Deficit	Success	Shots	Calls	Jobs	Capped target cost
Descriptor anchor only	0.38	72/72	0	0.0	0.0	0
GW anchor only	0.90	66/72	0	0.0	0.0	10923
Fixed donor	1.43	60/72	130560	64.0	23.0	21845
Random start	14.32	0/72	130400	65.0	23.2	131072
Descriptor, isotropic	0.38	72/72	130560	64.0	23.0	0
Descriptor, shaped	0.38	72/72	130560	64.0	23.0	0
GW, isotropic	0.90	66/72	130560	64.0	23.0	10923
GW, shaped	0.90	66/72	130560	64.0	23.0	10923
Local TV	1.47	60/72	130560	64.0	23.0	21845
Shuffled GW	1.51	54/72	130528	64.2	23.0	32768
Displaced prior	16.91	0/72	130432	65.6	23.4	131072
No point repair	0.38	72/72	130411	65.0	23.2	0
Fixed radius	0.38	72/72	129312	156.7	53.2	0
Fixed shots	0.38	72/72	66816	7.0	2.0	0
COBYLA	0.33	72/72	23282	22.7	22.7	0
SPSA	0.11	72/72	129024	126.0	84.0	0

between the Local TV and Fixed donor rows of Table 1 is an artifact of that tie-break and must not be read as evidence of structural selection. A bound of one cannot narrow the unit range of the normalized objective. Improved numerical conditioning alone is not evidence of saved shots, and a valid transfer bound can be uninformative on the evaluated graphs (Fig. 5). The results do not establish a consistent improvement from the complete transport-shaped policy over the simpler descriptor and conventional-optimizer controls. Comparisons with a tuned ASTRO-DF implementation and physical device noise remain outside this study.

Separately seeded panels test tighter targets and deeper circuits. The separate follow-up contains 448 test runs on 16 graphs from two separately seeded bank-validation-test panels, at depths 2 and 3. Its 128 tuning runs consume 8,170,240 optimization shots and 1,048,576 independent validation shots. The selected model-shot count and initial radius are frozen before any test run and shared by the isotropic and shaped variance-sensitive policies at each depth. All methods use a 65,536-shot online cap; each output receives 8,192 additional validation shots. Table 2 keeps initialization-only outcomes and actual online costs visible. Exact isomorphism checks identify 1 follow-up test graph also present in the canonical test set, leaving 15 structurally distinct test graphs. There is also 1 bank-to-bank overlap across the two studies. The follow-up bank, validation and test splits remain pairwise nonisomorphic within the new study. At depth 2, validation selects 256 model shots per point and initial radius 0.10. All four validation scores are tied exactly; the

Table 2: Separate follow-up test comparison. Each deficit is a mean signed reference gap in percentage points with a 95% graph-bootstrap interval, stratified within the two fixed reference banks (10,000 resamples; seed 5713). Each depth averages 16 graph means, with two shot repetitions per graph. Costs are mean online shots; add 8,192 final-validation shots per output and the separately reported tuning and bank costs.

Method	$p = 2$ deficit [95% interval]	$p = 3$ deficit [95% interval]	$p = 2$ shots	$p = 3$ shots
Descriptor anchor	0.41 [0.22, 0.64]	0.77 [0.44, 1.18]	0	0
Fixed schedule	2.94 [2.45, 3.46]	2.74 [2.31, 3.19]	0	0
Hoeffding, isotropic	0.41 [0.22, 0.64]	0.77 [0.44, 1.18]	64,768	64,512
Bernstein, isotropic	0.41 [0.22, 0.64]	0.73 [0.42, 1.14]	64,768	64,512
Bernstein, shaped	0.37 [0.22, 0.56]	0.66 [0.38, 1.01]	64,768	64,512
SPSA	0.21 [0.11, 0.35]	0.41 [0.20, 0.70]	64,512	64,512
COBYLA	0.32 [0.24, 0.41]	0.57 [0.37, 0.82]	22,720	30,752

first candidate in the declared order wins the tie. This outcome is not evidence of tuning superiority. Hoeffding, isotropic accepts 0 steps and changes 0/32 outputs, using 2,072,576 online shots in total. Bernstein, isotropic accepts 0 steps and changes 0/32 outputs, using 2,072,576 online shots in total. Bernstein, shaped accepts 1 step and changes 1/32 outputs, using 2,072,576 online shots in total. The paired expected-cut change from the isotropic to shaped variance-sensitive rule is 0.03 (95% interval 0.00 to 0.10) percentage points at depth 2; positive values favor shaping. At depth 3, validation selects 256 model shots per point and initial radius 0.10. All four validation scores are tied exactly; the first candidate in the declared order wins the tie. This outcome is not evidence of tuning superiority. Hoeffding, isotropic accepts 0 steps and changes 0/32 outputs, using 2,064,384 online shots in total. Bernstein, isotropic accepts 1 step and changes 1/32 outputs, using 2,064,384 online shots in total. Bernstein, shaped accepts 3 steps and changes 3/32 outputs, using 2,064,384 online shots in total. The paired expected-cut change from the isotropic to shaped variance-sensitive rule is 0.08 (95% interval 0.00 to 0.19) percentage points at depth 3; positive values favor shaping. These graph-level intervals do not estimate uncertainty over a population of reference banks. Separate panel means follow; each panel draws its bank, validation split, test split and random streams together, so a difference between panels is not attributable to the reference bank alone: panel 1, depth 2, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.46, 0.46, 0.46, 0.23 percentage points, respectively. panel 1, depth 3, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.99, 0.92, 0.77, 0.51 percentage points, respectively. panel 2, depth 2, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.35, 0.35, 0.29, 0.19 percentage points, respectively. panel 2, depth 3, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.55, 0.55, 0.55, 0.31 percentage points, respectively. At depth 2, the initialization-only anchor meets gaps 0.002, 0.005, 0.01 and 0.02 on 16/32, 26/32, 28/32, 32/32 test runs, respectively. Repeated initialization-only records correspond to the same anchor within a graph. At depth 3, the initialization-only anchor meets gaps 0.002, 0.005, 0.01 and 0.02 on 6/32, 18/32, 24/32, 28/32 test runs, respectively. Repeated initialization-only records correspond to the same anchor within a graph. An exclusion sensitivity analysis removes the overlapping graph without refitting or rerunning any method. On these 15 graphs at depth 2, the paired expected-cut change from isotropic to shaped Bernstein is 0.03 percentage points; the lowest mean deficits are SPSA (0.22) and COBYLA (0.33). The shaped rule has deficit 0.39, compared with 0.43 for initialization alone. On these 15 graphs at depth 3, the

Table 3: Transport sensitivity on 384 canonical graph pairs. “Unconv.” and “Infeas.” count selected pairwise solver candidates; changed donors and valid targets count graphs out of 24. Depth-specific deficits are mean anchor-only signed reference gaps in percentage points over valid targets. The first two rows are reference points rather than audit solvers: the study’s own capped entropic solver and the transport-free descriptor selector, both over all 24 targets. Feasibility, convergence, and useful donor selection are distinct properties.

Solver	Unconv.	Infeas.	Changed donors	Valid targets	$p = 1$ deficit	$p = 2$ deficit
Original capped entropic (study)	253	0	–	24	0.29	0.90
Descriptor selector (no transport)	–	–	–	24	0.09	0.38
Entropic $\epsilon = 0.02$	100	99	1	1	0.15	0.50
Entropic $\epsilon = 0.05$	3	0	1	24	0.30	0.89
Entropic $\epsilon = 0.1$	5	0	20	24	0.31	1.01
Entropic, three starts	14	0	10	24	0.23	0.74
Conditional gradient, three starts	0	0	21	24	0.24	0.65

paired expected-cut change from isotropic to shaped Bernstein is 0.08 percentage points; the lowest mean deficits are SPSA (0.43) and COBYLA (0.58). The shaped rule has deficit 0.67, compared with 0.78 for initialization alone. The held-out test runs consume 18,250,752 optimization shots and 3,670,016 additional validation shots. All follow-up best-found references together use 97,183 exact objective calls, including 40,409 bank-reference calls; these are classical simulation costs, not uncharged hardware shots. The added panels enlarge the tested regime but remain ideal-circuit, small-graph experiments with a fixed, disclosed tuning grid.

Solver sensitivity limits structural-selection conclusions. The separate transport audit repeats every canonical target–reference comparison with longer iterations, three regularizations, multiple starts, and independent conditional-gradient optimization. All saved couplings are checked by a direct four-index distortion calculation. Conditional-gradient stopping is checked by a transportation linear-program gap; this is a first-order stationarity diagnostic at the stated numerical tolerance, not a certificate of exact stationarity or global optimality. Table 3 reports solver failures alongside changed donors and resulting anchor quality. A target with any infeasible candidate is excluded from that selector’s quality summary and explicitly counted; no finite score is substituted for an invalid solve. Quality means over different valid-target subsets are not paired policy comparisons and must not be used to rank those selectors directly.

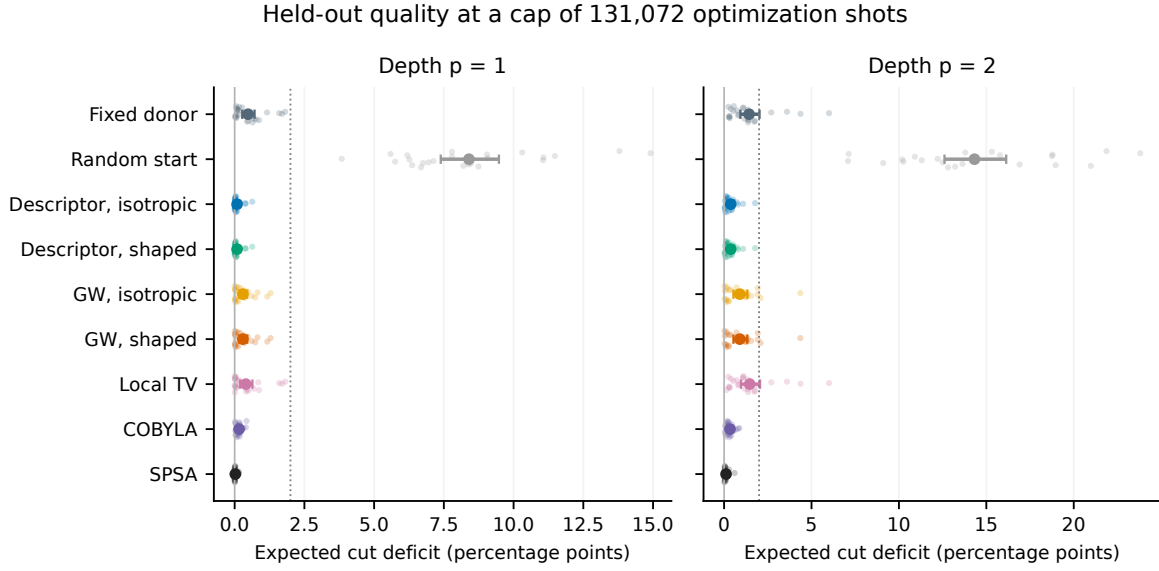


Figure 2: Final quality at a 131072-shot optimization cap, at depths one and two. Deficit is 100 times the signed gap between the returned normalized objective and the best-found reference; smaller is better. The dotted line marks the target gap of 0.02. Points are the 24 graph means after averaging three shot seeds. Intervals use 10000 graph-bootstrap resamples (95% percentile). Exact objectives are diagnostic; independent final validation is outside the cap.

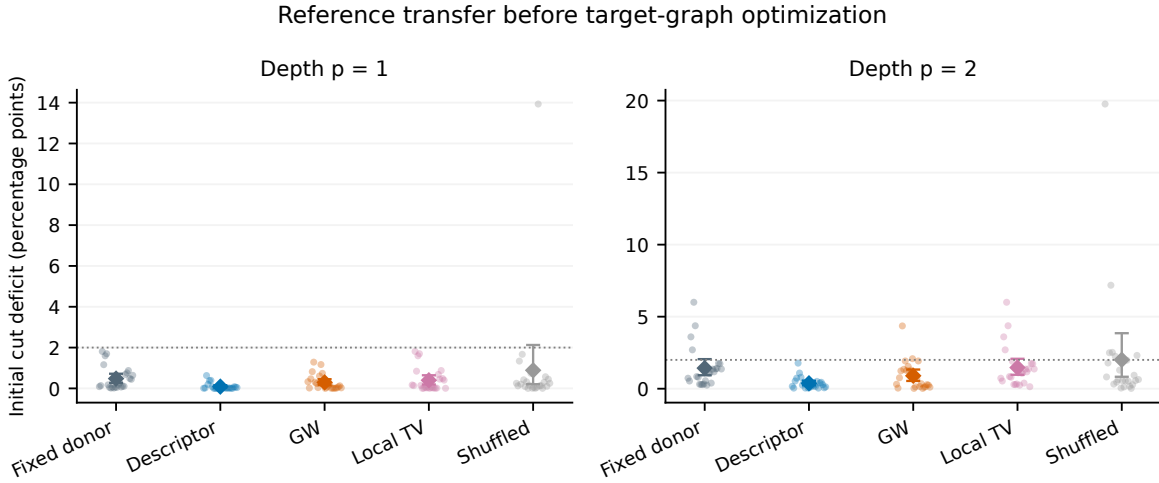


Figure 3: Initial transfer deficits, in percentage points, before target-graph refinement. Points are deterministic initial deficits on 24 graphs; intervals use 10000 graph-bootstrap resamples (95% percentile). Vertical scales differ between depths. Best-found target references are used only for evaluation. Theorem 1 applies to rooted-neighborhood total variation; the other selectors are heuristic. All depth-two local scores mathematically equal one: the separation of Fixed donor and Local TV comes from floating-point selection, not graph structure.

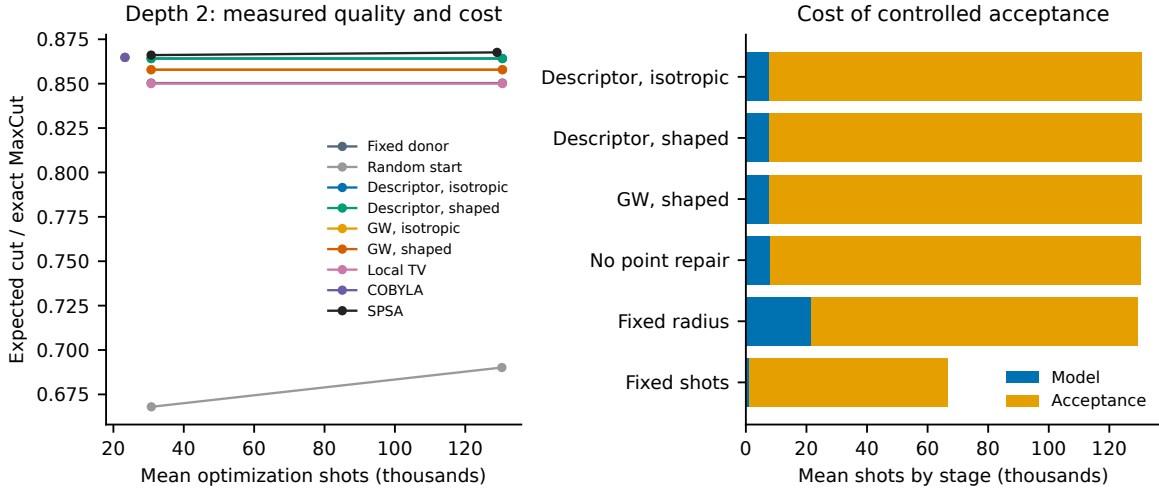


Figure 4: Measurement cost and quality at depth two. The quality–cost panel plots expected cut divided by exact MaxCut against actual mean optimization shots at the two budget caps. The cost decomposition separates model fitting from acceptance measurements for six local-model variants. Rejected and unresolved trials remain in the accounting. The common final-validation cost of 16384 shots per run is additional.

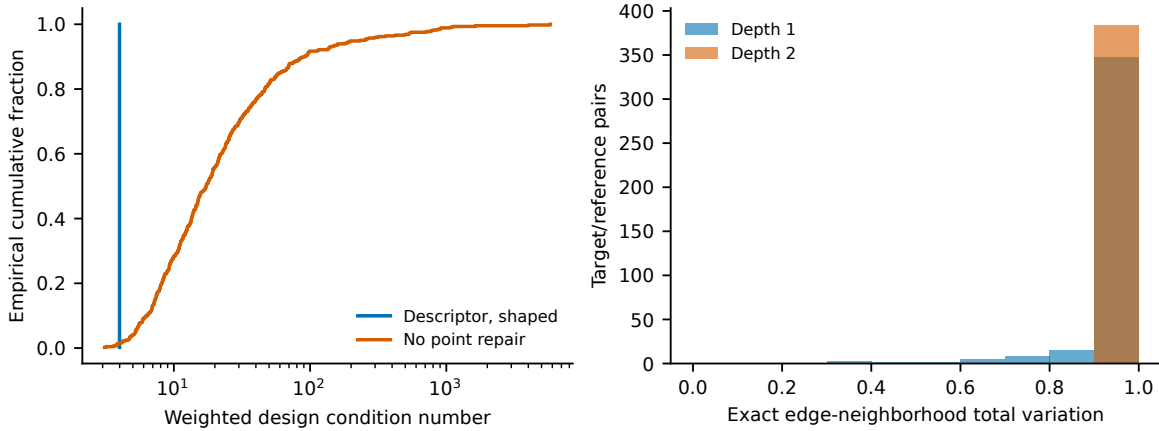


Figure 5: Numerical geometry and the structural certificate. The empirical distribution of weighted-design condition numbers compares QR-selected and unselected random interpolation sets at depth two. The histograms show exact rooted-neighborhood total variation for target–bank graph pairs at depths one and two. A value of one gives a vacuous normalized objective bound; at depth two every pair attains that value, so the depth-two histogram is a single bar and the score cannot order the reference bank. Numerical conditioning and structural matching describe different parts of the method and do not alone establish shot savings.

3 Discussion

The central empirical result is the separation between a useful transferred anchor and useful finite-shot refinement. In the canonical study, descriptor initialization meets the target on every test graph, and the nine undisplaced donor-based local-model policies retain their anchors. Their final quality is therefore entirely attributable to transfer. Equal quality from isotropic and shaped regions in these runs follows from equal returned vectors; it cannot establish that the two geometries are equally effective when refinement occurs. The supported practical choice for this sample is descriptor initialization, with any subsequent optimizer assessed against that initialization-only control.

Statistical resolution gives this result a quantitative interpretation. For the prescribed Hoeffding rule, a trial whose true decrease exceeds the required model decrease by at most 0.005 has conditional acceptance probability below 3.5×10^{-6} across all four canonical looks (Proposition 5). This bound applies even when the last look is affordable; a smaller budget can only remove opportunities for acceptance. Conversely, the sufficient detection condition scales inversely with the square of that margin (Corollary 4). These are properties of the stated decision policy, not universal measurement lower bounds for QAOA. Together with the model-error relation, they explain why improving interpolation conditioning alone need not produce a measurable refinement benefit.

The follow-up changes the decision rule while keeping matched isotropic models and initializations. Empirical-Bernstein acceptance resolves one isotropic move where the corresponding Hoeffding policy resolves none. The shaped Bernstein variant resolves four moves and lowers the mean initialization deficit by about 0.03 and 0.11 percentage points at depths two and three. No shaped Hoeffding cell is run, so the shape-by-rule interaction remains unidentified. SPSA and COBYLA achieve lower mean deficits at both depths. Moreover, every validation candidate returns its anchor and has the same validation score: the selected radius and model shot count result from the declared tie-break, with no demonstrated tuning advantage. The follow-up establishes that a few improvements become resolvable under variance-sensitive acceptance, while leaving the overall optimizer ranking unfavorable to the local-model policies.

Structural selection has a separate limitation. The rooted-neighborhood certificate is vacuous on 301 of 384 canonical graph pairs at depth one and on all 384 at depth two. At depth two, the Local TV donor differs from the fixed donor only because rounding in mathematically tied scores changes three selections. The shortest-path transport score also fails to improve on the descriptor selector in this sample. Its original finite-iteration solver misses the outer tolerance on 253 pairs, including 20 selected donors. Longer entropic solves change only one donor, while independent conditional-gradient solves change 21 and reduce the mean depth-two transfer deficit to about 0.65 percentage points, still above the descriptor value of 0.38. The lowest regularization leaves only one target with a complete feasible donor comparison set, preventing comparison of its conditional mean with the 24-target means. Solver feasibility, stopping criteria, and transfer quality therefore require separate assessment.

The region’s reference scatter is a geometric prior, without a calibrated probabilistic interpretation. Accepted moves translate the region and measurement decisions change its radius; the deliberately displaced prior tests one inaccurate-initialization scenario. These mechanisms do not prove recovery from every poor prior. Fixed model shots, a positive minimum radius, and a finite horizon also preclude importing asymptotic stochastic trust-region convergence guarantees. A time-uniform confidence sequence [19] could replace the prescribed within-trial looks, but would require a suitable predictable variance construction and an error allocation across adaptive trials. Its efficiency remains to be tested.

The evidence concerns ideal-circuit sampling on small graphs. Gate and readout errors, calibra-

tion drift, routing, communication latency, and hardware scheduling are absent. Logical batches and simulator timings do not supply hardware runtime estimates. Offline exact optimization supplies the donor bank at a classical cost that limits scaling; deployment comparisons must include this cost and specify its amortization. The measured baselines are SPSA, COBYLA, and the stated ablations, rather than a comprehensive ranking against the model-based, gradient-based, or graph-learning methods discussed in the Introduction. Their training and measurement costs, as well as stronger baseline tuning, remain relevant comparisons.

The canonical study has 24 test graphs and one bank; the follow-up has 16 test graphs and two separately seeded banks. One follow-up test graph is isomorphic to a canonical test graph; exclusion sensitivity retains the reported ordering. Repeated shot seeds do not increase the independent graph count, and two additional banks provide limited information about bank variability. The follow-up was designed after the canonical results, with its own validation choices fixed before test measurements. Numerical backend changes can also select different nearly tied reference solutions or finite-iteration GW candidates, so the saved binary-environment record is part of reproducibility. These restrictions define the scope of the findings: initialization dominates in the evaluated regime, and a stronger refinement claim requires a demonstrable match between available improvement, model accuracy, and measurement resolution.

4 Methods

4.1 QAOA objective, symmetries, and measurements

For an unweighted simple graph $G = (V, E)$ with $n = |V|$ vertices (one qubit each) and $m = |E| > 0$ edges, the cost and mixer Hamiltonians are

$$C_G = \sum_{(i,j) \in E} \frac{I - Z_i Z_j}{2}, \quad H_X = \sum_{i \in V} X_i. \quad (1)$$

With angles ordered as $\theta = (\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p)$, we use

$$|\psi_G(\theta)\rangle = e^{-i\beta_p H_X} e^{-i\gamma_p C_G} \dots e^{-i\beta_1 H_X} e^{-i\gamma_1 C_G} |+\rangle^{\otimes n}. \quad (2)$$

The minimization objective is $f_G(\theta) = -\langle \psi_G(\theta) | C_G | \psi_G(\theta) \rangle / m$. A computational-basis measurement z produces one observation $Y = -C_G(z)/m \in [-1, 0]$. Its mean is $f_G(\theta)$. The simulator samples the exact cut-value distribution by a multinomial draw; it does not replace the shot distribution by additive Gaussian noise.

Each γ has period 2π because the unweighted cut spectrum is integer. Each β has period $\pi/2$ in the objective because the associated global bit flip commutes with every cost layer and preserves the initial state up to phase. Simultaneously reversing all angles complex-conjugates the state and preserves the objective. Reference vectors are aligned to an anchor using these symmetries: for each of the two global sign choices, every coordinate is shifted by its period to the representative nearest the anchor, and the alignment with the smaller Euclidean distance to the anchor is retained. The procedure does not assume that every symmetry-equivalent optimum has been found.

All optimization decisions use shot estimates. Exact expectations are computed only for reference construction and recorded diagnostics, including post hoc objective trajectories. The reported expected approximation ratio is $F_G(\theta)/C_{\max}(G)$, with $C_{\max}$ obtained by exhaustive bitstring enumeration at these graph sizes. It is distinct from the largest cut observed among a collection of samples. A single depth- p circuit uses pm cost-edge rotations and pn mixer rotations; a two-CNOT implementation of each cost-edge rotation would use $2pm$ CNOTs per shot, before routing and compilation. We do not convert this logical count into a hardware runtime claim.

4.2 A structural certificate for objective transfer

For a graph $G = (V, E)$ with $m = |E| > 0$, let $F_G(\theta)$ be the expected MaxCut value at depth p , and define the minimized objective $f_G(\theta) = -F_G(\theta)/m \in [-1, 0]$. For an edge e , form the induced subgraph containing every vertex within distance p of either endpoint, and mark the endpoints as an unordered root pair. Write $\mu_{G,p}$ for the distribution of rooted isomorphism types obtained by sampling an edge uniformly from E .

The causal structure of QAOA yields the following uniform comparison.

Theorem 1 (Transfer through rooted edge neighborhoods). *For any two finite simple unweighted graphs with at least one edge, and any shared depth p , define $d_p(G, H) = \text{TV}(\mu_{G,p}, \mu_{H,p}) = \frac{1}{2} \sum_{\tau} |\mu_{G,p}(\tau) - \mu_{H,p}(\tau)|$, total variation in the convention that carries the factor one half, so that $d_p \in [0, 1]$ and $d_p = 1$ exactly when the two type distributions have disjoint supports. Then*

$$\sup_{\theta \in \mathbb{R}^{2p}} |f_G(\theta) - f_H(\theta)| \leq d_p(G, H). \quad (3)$$

If θ_H is ϵ -optimal for f_H on a common nonempty domain Ω , then

$$f_G(\theta_H) - \inf_{\theta \in \Omega} f_G(\theta) \leq \epsilon + 2d_p(G, H). \quad (4)$$

The proof below uses the established locality of fixed-depth QAOA edge expectations [2, 11]; related local-subgraph comparisons motivate parameter-transfer heuristics [12]. The formulation here makes the finite-graph total-variation and donor-regret consequences explicit. It gives a sufficient transfer certificate without requiring equal graph sizes or a vertex correspondence. It is conservative: different neighborhoods can produce the same edge expectation, and disjoint type distributions give $d_p = 1$, for which Eq. (3) gives no information beyond $f_G \in [-1, 0]$. The factor-two regret bound can be clipped at one. Equation (4) is additionally conditional on the donor: it presumes a value of ϵ for which θ_H is ϵ -optimal for f_H on Ω . The donor angles used here are the lowest objective among six BFGS runs on the exact expectation, a multistart local minimization that certifies no optimality gap, so no numerical ϵ is available and Eq. (4) is never evaluated for a graph pair in the empirical sections. Computing exact rooted isomorphism classes can also be expensive.

The global shortest-path score used in the GW construction does not appear in Eq. (3). The two objects are not measures on a common space: the GW coupling compares global shortest-path relations under uniform *vertex* masses $a_i = 1/|V_G|$, whereas d_p compares rooted local neighborhood types under uniform *edge* sampling, and no theorem here connects a small GW distortion to a small d_p . Small GW distortion is therefore a heuristic reason to try transfer, not a certificate of objective agreement. Even a structural comparator with an exact certificate can be uninformative on a particular finite sample. The empirical comparison evaluates both the quality of the selected reference and the informativeness of the scores.

Proof of Theorem 1. For a fixed edge $e = (u, v)$, let $C_e = (I - Z_u Z_v)/2$. Propagate C_e backwards through Eq. (2). A mixer layer acts separately on qubits and cannot enlarge its support. Within a cost layer all ZZ gates commute. Gates with neither endpoint in the current support commute with the operator and cancel, while the remaining gates can add only immediate neighbors. Thus each cost layer increases the maximum graph distance from the original edge by at most one. After p layers, only the induced radius- p edge neighborhood can affect the expectation. Extra edges entirely on its outer boundary cause no problem: they are present in the induced graph, and gates that never meet the backwards support cancel.

The product initial state and the uniform circuit angles imply that root-preserving isomorphic neighborhoods have identical edge expectations. Hence, for a rooted type τ , there is a common function $h_p(\tau, \theta) \in [0, 1]$ such that

$$-f_G(\theta) = \sum_{\tau} \mu_{G,p}(\tau) h_p(\tau, \theta). \quad (5)$$

For any function taking values in $[0, 1]$, the difference of its expectations under two distributions is at most their total variation. This proves Eq. (3), uniformly in θ . Furthermore, $f_G(\theta_H) \leq f_H(\theta_H) + d_p$ and $\inf_{\Omega} f_H \leq \inf_{\Omega} f_G + d_p$. Combining these inequalities with the assumed ϵ -optimality proves Eq. (4). $\square$

The implementation counts exact rooted isomorphism classes. Degree and size invariants are used only to reject impossible matches, followed by an exact graph-isomorphism test with endpoint marks. No hash equality is treated as an isomorphism certificate.

4.3 Structural comparison and reference selection

For connected graphs, define $D_G(i, k) = \text{dist}_G(i, k) / \text{diam}(G)$ and uniform vertex weights $a_i = 1/|V_G|$, $b_j = 1/|V_H|$. A feasible coupling satisfies $T \geq 0$, $T\mathbf{1} = a$, and $T^\top \mathbf{1} = b$. Its squared structural distortion [24, 28] is

$$\mathcal{D}_{G,H}(T) = \sum_{i,k,j,\ell} (D_G(i, k) - D_H(j, \ell))^2 T_{ij} T_{k\ell}. \quad (6)$$

Following the regularized structural-comparison idea in Refs. [7, 28], the independent implementation starts from $T = ab^\top$ and applies damped entropic transport updates with regularization 0.05. With $K_{ij} = (D_G^{\circ 2} a)_i + (D_H^{\circ 2} b)_j - 2(D_G T D_H^\top)_{ij}$, the next proposal is obtained by Sinkhorn scaling the kernel $\exp(-2K/0.05)$, following the fixed-cost entropic transport construction [6]. The actual update averages the proposal and previous coupling with equal weights. The score reported is the *unregularized* distortion of the returned coupling.

The outer loop stops at a coupling-change Frobenius norm below 10^{-9} or after 100 updates. Log-domain inner scaling uses at most 500 iterations and a row-marginal tolerance of 10^{-12} ; both marginals are checked to 10^{-8} before accepting a returned coupling. The tensor K is rounded to 12 decimal places before forming the kernel. This fixed numerical guard suppresses roundoff-scale symmetry breaking at the uniform start. It does not certify a stationary point, a global optimum, or exact invariance under all floating-point perturbations. Uniform initialization can also remain uninformative on symmetric instances. Couplings, residuals, iteration counts, and marginal errors are retained. In the canonical study, 253 of 384 distinct target–bank comparisons terminate without reaching the 10^{-9} outer tolerance. Twenty of the 24 selected donors have such a capped comparison. Scores are reused across depths because they depend only on the two graphs. The empirical result therefore concerns this capped, regularized solver with uniform initialization; it does not establish the quality of globally optimized GW matching.

The inexpensive alternative is a six-component descriptor: mean and standard deviation of degree, its quartiles at 0.25, 0.5, and 0.75, and mean clustering coefficient. Descriptor differences are scaled by their reference-bank standard deviations, floored at 0.1, and compared in Euclidean norm. The local alternative uses d_p from Theorem 1. Neither reference choice uses the target graph’s optimized parameter label.

For scores d_j to bank graphs, let

$$\begin{aligned} w_j &= \frac{\exp(-d_j/\tau)}{\sum_k \exp(-d_k/\tau)}, \\ \tau &= \max\{\text{sd}(d_j), 10^{-3}\}. \end{aligned} \tag{7}$$

The code subtracts the minimum score before exponentiation for numerical stability; this leaves the normalized weights unchanged. The initial anchor x_0 is the first largest computed weight in the saved bank order. In exact arithmetic the weights decrease strictly with the scores, so this selects a score minimizer; in floating point, selection uses the computed weights without a near-tie tolerance. Rounding-scale score differences can therefore change the selected donor. For the local alternative at depth two, all 384 target–bank scores equal one to within 4×10^{-16} , so Eq. (3) is vacuous for every pair. The selected reference is nevertheless not the first bank vector on 3 of the 24 depth-two targets, and those 3 targets are the only source of the difference between the local-score and fixed-donor rows of Table 1; that difference is a rounding artifact and carries no structural information. In particular, the fixed-donor control uses the first bank vector rather than an average. After symmetry alignment, only reference vectors θ_j satisfying $\|\theta_j - x_0\|_2 \leq 1$ are retained for the shape, and their weights are renormalized. Define

$$V = 0.02I + \sum_j w_j (\theta_j - x_0)(\theta_j - x_0)^\top. \tag{8}$$

Subtract the mean log eigenvalue of V , clip each resulting value to $[-\log 2, \log 2]$, and subtract their mean once more. If the resulting values are ℓ_j and the eigenvectors form U , set $B = U \text{diag}(e^{\ell_j/2}) U^\top$. Then $\det B = 1$ and $\text{cond}(BB^\top) \leq 4$. Both follow from the two recentrings and the clip: the final subtraction of the mean gives $\sum_j \ell_j = 0$, so $\det B = \exp(\frac{1}{2} \sum_j \ell_j) = 1$; the clip confines the pre-centred values to an interval of length $2 \log 2$, and subtracting a constant leaves that spread unchanged, so $\max_j \ell_j - \min_j \ell_j \leq 2 \log 2$. The eigenvalues of $BB^\top$ are e^{ℓ_j} , whose ratio is therefore at most $e^{2 \log 2} = 4$. This scatter matrix is an anchor-centred scatter of the retained reference vectors, not a covariance about their weighted mean, and is used only for geometry, with no probabilistic interpretation.

4.4 Sensitivity of the transport selector

The transport sensitivity study reuses all 384 canonical target–bank pairs and their saved parameter references. It changes structural computation only: no target optimizer is rerun, and target QAOA objective values never select a solver or a start. For entropic updates, the outer limit becomes 1000 while the coupling-change tolerance remains 10^{-9} . The inner loop retains 500 scaling iterations, tolerance 10^{-12} , and the final marginal tolerance 10^{-8} . Uniform-start regularizations are 0.02, 0.05, and 0.1. A separate 0.05 multistart policy adds two seeded random positive couplings, balanced to uniform marginals, and retains the feasible returned coupling with lowest unregularized distortion. The protocol fixes seed 2026091903 and deterministic offsets by graph pair and start. The rounded cost tensor uses 12 decimal places as before. Positive-kernel scaling recenters exponents by their maximum; comparison with the original log-domain implementation is tested independently.

An additional solver applies the nonconvex conditional-gradient construction of Ref. [22] to the unregularized objective in Eq. (6). At a feasible coupling T , the nonconstant part of its gradient is $-4D_G T D_H^\top$. Each search direction solves a transportation linear program with uniform marginals using SciPy’s HiGHS interface at primal and dual feasibility tolerances of 10^{-9} ; the omitted separable gradient terms are constant along feasible directions. The step minimizes the quadratic

objective along the feasible segment by comparing both endpoints and any interior minimizer. The solver uses the same three starts, at most 200 updates, and a final first-order gap tolerance of 10^{-10} , the gap being $\langle -4D_G T D_H^\top, T - S \rangle$ with S a minimizer of the same linear form over feasible couplings. The lowest-distortion returned coupling defines the comparison score. A small first-order gap is a stationarity diagnostic, not a global-optimality certificate for this nonconvex problem. On 48 of the 384 pairs the uniform start already meets the gap tolerance without any update, because two bank graphs have constant normalized distance row sums; in every such pair a random start returns strictly lower distortion. Numerically solved linear subproblems and finite precision make the reported gap a floating-point diagnostic rather than an exact stationarity certificate. The 10^{-10} gap threshold also lies below the 10^{-9} feasibility tolerance of those subproblems.

Every start’s coupling, distortion, residual or gap, iteration count, feasibility status, and marginal error is retained, including finite infeasible entropic results. The conditional-gradient routine instead raises on a returned coupling that violates the marginal tolerance, so its recorded couplings are feasible by construction rather than by observation. A target has a valid selector only when all 16 donor comparisons have feasible returned couplings; otherwise its result is recorded as unavailable. The rule does not silently discard difficult donors. Reported transfer means condition on the valid targets, whose counts are shown in Table 3; different valid-target subsets cannot support a direct ranking. For feasible targets, the lowest structural score selects a donor in fixed bank order. Its saved angles are evaluated on the target only to diagnose transfer quality. Direct four-index summation independently checks each saved distortion, and a fresh transportation subproblem checks the conditional-gradient gap. These checks support the recorded finite computations without upgrading them to exact GW optima.

4.5 Geometry controls local-model noise amplification

At a current angle vector x , write the trial region as

$$\begin{aligned} \mathcal{T}(x, \Delta, B) &= \{x + \Delta B u : \|u\|_2 \leq 1\}, \\ B &= B^\top \succ 0, \quad \det B = 1. \end{aligned} \tag{9}$$

The equivalent metric is $M = B^{-2}$. The shape B is fixed during a run; the center and radius change in response to measurements. The determinant constraint preserves volume when comparing shapes at the same radius. Angles are evaluated modulo exact QAOA symmetries; local models are interpreted in the unwrapped coordinates of Eq. (9).

For local coordinates u_i , the affine design has rows $A_i = (1, u_i^\top)$. The model $m(u) = \hat{c}_0 + \hat{c}_{1:d}^\top u$, $d = 2p$, is fitted from shot estimates by weighted QR. The implementation always evaluates exactly $N = d + 1$ points at a common shot count, so A is square and invertible and the weighted fit reduces to interpolation of the sample means: any positive weights leave $\hat{c}$ unchanged, and no accuracy benefit is claimed for weighting. The weights are retained only because Proposition 2 is stated for $N \geq d + 1$. If $W^{1/2}A = QR$ has full column rank, then

$$\hat{c} - c^* = R^{-1}Q^\top W^{1/2}(r + e), \tag{10}$$

where c^* is the local first-order Taylor model, r is its bias at the sample points, and e is measurement error. Proposition 2 bounds each contribution under explicit smoothness and sampling conditions. The smallest singular value of R determines how strongly the fitting geometry can amplify these errors.

The implementation selects $d+1$ interpolation points by ordinary column-pivoted QR [3] applied to the transpose of a candidate design. All candidate coordinates and basis evaluations are classical;

only the selected points incur measurements. This procedure preserves full model dimension. It does not perform adaptive low-rank approximation and does not claim a new QR factorization. Comparing with an unselected random design tests the value of point placement while holding the least-squares solver fixed. Ordinary pivoting alone supplies neither a strong rank-revealing guarantee nor an optimal interpolation set.

Point selection and local-model error. For dimension $d = 2p$, draw $d + 1$ independent standard Gaussian vectors and normalize each to the unit sphere. The repaired candidate set contains these vectors, 0, and the signed coordinate vectors $\pm e_j$. Column-pivoted QR of the transposed candidate design selects $d + 1$ rows. The no-repair variant uses the original random sphere points. Both check full rank and use the same QR solver. No measured objective enters point selection. The implementation records the condition number of the weighted triangular factor, not the smallest singular value $\sigma_{\min}(R)$ that Eq. (13) inverts; a condition number alone does not bound the inverse-design sensitivity. For the repaired design the pivoted selection returned the same point set in every recorded trial, so this quantity is in fact constant: at the canonical 256 shots per point, $\sigma_{\min}(R) = 12.25$ at $p = 1$ and 9.52 at $p = 2$, while the recorded condition number is correspondingly fixed at $1 + \sqrt{2} \approx 2.414$ and 3.992. With $\delta = 0.05$ the stochastic term of $\mathcal{E}$ is then about 0.22 and 0.38, which is not small against the unit range of the normalized objective. Interpolation geometry and inverse-design bounds have an established role in local-model accuracy [5, Sections 3–5]. The following proposition combines a Taylor remainder, weighted least-squares perturbation, and bounded-observation concentration for the specified noisy affine model; full rank alone does not impose all conditions of a fully linear model.

Proposition 2 (Local model error with bounded observations). *Suppose ∇f is L_f -Lipschitz on a convex set containing $\mathcal{T}(x, \Delta, B)$. Choose $N \geq d + 1$ points with $\|u_i\|_2 \leq 1$ and positive shot counts S_i before drawing the stage’s independent observations in $[-1, 0]$. Let $W = \text{diag}(S_i)$ and let $W^{1/2}A = QR$ have full column rank. Define*

$$c^* = (f(x), \Delta B^\top \nabla f(x))^\top, \quad (11)$$

$$a_i = \frac{1}{2} L_f \Delta^2 \|Bu_i\|_2^2, \quad \sigma = \sigma_{\min}(R) > 0. \quad (12)$$

For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\|\hat{c} - c^*\|_2 \leq \frac{\sqrt{\sum_i S_i a_i^2} + \sqrt{\frac{N}{2} \log(2N/\delta)}}{\sigma} =: \mathcal{E}. \quad (13)$$

On the same event, uniformly for $\|u\|_2 \leq 1$,

$$|f(x + \Delta Bu) - m(u)| \leq \frac{1}{2} L_f \Delta^2 \|B\|_2^2 + \sqrt{2} \mathcal{E}. \quad (14)$$

These statements also hold conditionally when the design and shot counts are chosen from the history preceding this stage.

Proof. The first-order Taylor remainder satisfies $|r_i| \leq a_i$. Writing the vector of sample means as $y = Ac^* + r + e$ gives Eq. (10), since $\hat{c} = R^{-1}Q^\top W^{1/2}y$ and $Q^\top Q = I$. Hoeffding’s inequality [18] and a union bound give

$$|e_i| \leq \sqrt{\frac{\log(2N/\delta)}{2S_i}} \quad \text{for every } i \quad (15)$$

with probability at least $1 - \delta$. Therefore $\|W^{1/2}r\|_2 \leq \sqrt{\sum_i S_i a_i^2}$ and $\|W^{1/2}e\|_2 \leq \sqrt{N \log(2N/\delta)/2}$. Using $\|R^{-1}Q^\top\|_2 = 1/\sigma$ proves Eq. (13). At any point in the region, the Taylor remainder is at most $L_f \Delta^2 \|B\|_2^2/2$ and $\|(1, u^\top)\|_2 \leq \sqrt{2}$. The triangle inequality proves Eq. (14). Conditioning fixes all choices made before the independent stage observations, so the same argument applies to a history-dependent design. $\square$

For QAOA the objective is analytic, so the required smoothness holds on any compact region, but the theorem does not supply a small useful L_f . Equal shot counts S make the stochastic term proportional to $S^{-1/2}/\sigma_{\min}(A)$; the bias term need not decrease with S . Because the gradient block of c^* is $\Delta B^\top \nabla f(x)$, the recovered gradient is $\Delta^{-1} B^{-\top} \hat{c}_{1:d}$, whose stochastic error scales as $\Delta^{-1} S^{-1/2}$. Contracting the radius from the canonical 0.4 to the floor 0.025 is a factor of 16, so 256 times the shots per point would preserve this stochastic gradient-error bound at the same design and shape. In the recorded runs 26.2% of radius-adapting trials are taken at the floor radius, and none of the 240 accepted steps occurs at a radius below 0.15. The implemented model uses $S = 256$ at each selected point at every radius and does not estimate L_f or choose S by enforcing Eq. (14). Consequently, the proposition explains sources of model error without certifying every fitted model as sufficiently accurate for convergence.

Equation (14) also links model quality to the acceptance rule of the next subsection. Write $\mathcal{E}_f = \frac{1}{2} L_f \Delta^2 \|B\|_2^2 + \sqrt{2} \mathcal{E}$ for the uniform model error, let $q = \|\hat{c}_{1:d}\|_2$ be the decrease predicted by the model's boundary step, and let a be the true decrease at that step. Applying Eq. (14) at the center and at the proposed point gives $a \geq q - 2\mathcal{E}_f$, so the detection condition $a - \eta q \geq 4b_L$ of Corollary 4, with acceptance slack $\eta \in (0, 1)$ and confidence allowance b_L from Eq. (17), is implied by

$$(1 - \eta)q \geq 2\mathcal{E}_f + 4b_L. \quad (16)$$

Provided the last prescribed look is affordable, this is a sufficient joint condition on proposal quality and measurement resolution, on the intersection of the model and acceptance confidence events. Failure of the condition does not rule out a useful or accepted step: the error bounds may be conservative. Neither $\mathcal{E}_f$ nor L_f is estimated in the runs, so Eq. (16) provides an analytical connection rather than an implemented test.

4.6 Accepted steps have a finite-horizon statistical guarantee

The affine model proposes the boundary step $u^+ = -\hat{c}_{1:d}/\|\hat{c}_{1:d}\|_2$ when its gradient is nonzero. Its predicted decrease is $q = m(0) - m(u^+) = \|\hat{c}_{1:d}\|_2 > 0$. Fresh measurement streams estimate the actual decrease $a = f_G(x) - f_G(x + \Delta B u^+)$. At prespecified cumulative sample sizes $s_1 < \dots < s_L$, define

$$b_\ell = \sqrt{\frac{\log(4TL/\alpha)}{2s_\ell}}, \quad \hat{a}_\ell = \bar{Y}_{x,\ell} - \bar{Y}_{+, \ell}, \quad (17)$$

where T bounds the number of trials and α is the failure probability for one optimization run. A trial is accepted only if

$$\hat{a}_\ell - 2b_\ell \geq \eta q, \quad 0 < \eta < 1. \quad (18)$$

Theorem 3 shows that, with probability at least $1 - \alpha$, every accepted step in that run satisfies $a \geq \eta q > 0$, despite adaptive trial points and early stopping among the prescribed looks. Model accuracy is not needed for this acceptance statement, because acceptance uses fresh bounded observations.

This separation is useful but has a cost. A poor model can propose many unacceptable steps, and a small true improvement can remain unresolved within the allowed shots. A trial whose

interval still intersects the threshold at the last affordable look is recorded as unresolved and the incumbent is retained. Such a decision is not evidence that the trial is worse. With $T = 64$, $L = 4$, $\alpha = 0.05$, and the final look $s_L = 32768$, the two-endpoint confidence allowance $2b_L$ is approximately 0.0246 in normalized objective units. Reaching that look costs 65536 shots for the endpoint pair, so it is unaffordable at the 32768-shot budget; in the canonical study it is reached by 1,440 of the 21,196 trials that follow the four prescribed looks, all at the larger budget, while the deepest look reached at the smaller budget is 8192 shots per endpoint, where the allowance $2b_\ell$ is approximately 0.0492. This statistical scale must be considered when interpreting small expected improvements.

Simultaneous validity and detectable improvements.

Theorem 3 (Simultaneous validity over prescribed looks). *Consider at most T adaptive trials. Before a trial's acceptance measurements, its center, trial point, and predicted decrease $q > 0$ are fixed conditional on the previous history. At each endpoint, start a fresh stream of conditionally independent observations in $[-1, 0]$ with the corresponding true objective mean. Use only the L cumulative sample sizes specified in advance and the bounds in Eq. (17). With probability at least $1 - \alpha$, all endpoint confidence intervals at all executed trials and looks are valid. On this event every acceptance by Eq. (18) satisfies*

$$f_G(x) - f_G(x + \Delta Bu^+) \geq \eta q > 0. \quad (19)$$

Early stopping or budget-based omission of a prescribed look does not change the guarantee.

Proof. Condition on the history before one trial. For a fixed endpoint and look, Hoeffding's inequality gives error probability at most $2\exp(-2s_\ell b_\ell^2) = \alpha/(2TL)$. A union bound over two endpoints and L looks bounds that trial's conditional failure probability by α/T . Taking expectations and union bounding over at most T trials gives a total failure probability at most α . The argument uses fresh streams after trial selection; within a trial, cumulative estimates may share observations. On simultaneous validity, $|\hat{a}_\ell - a| \leq 2b_\ell$, so Eq. (18) implies $a \geq \eta q$. Stopping selects a subset of the controlled events and therefore cannot increase this probability bound. $\square$

Corollary 4 (A sufficient measurement scale for detection). *Under the assumptions of Theorem 3, fix a trial with true decrease a and predicted decrease q . If its last prescribed look is affordable and*

$$a - \eta q \geq 4b_L > 0, \quad (20)$$

then on the simultaneous confidence event it is accepted at or before that look. Equivalently, for a margin $h = a - \eta q > 0$, the sufficient final sample count per endpoint is

$$s_L \geq \frac{8 \log(4TL/\alpha)}{h^2}. \quad (21)$$

Proof. On the stated event $|\hat{a}_\ell - a| \leq 2b_\ell$ at every look. No earlier rejection is possible when $a > \eta q$, since $\hat{a}_\ell + 2b_\ell \geq a > \eta q$. At the final look, $\hat{a}_L - 2b_L \geq a - 4b_L \geq \eta q$, which gives acceptance. Substituting Eq. (17) and squaring positive quantities proves Eq. (21). $\square$

The sufficient margin is about 0.0492 for the canonical final look. Even if every observed mean equalled its expectation, the canonical rule would still require $a - \eta q \geq 2b_L \simeq 0.0246$ at that look. This last statement describes the decision threshold in the absence of sampling error, not an impossibility bound for random observations. The corollary is a detection guarantee conditional on

a sufficiently good proposal; it does not ensure that the local model produces one. Its inverse-square dependence on the improvement margin helps explain why a statistically valid rule can spend a large budget without certifying useful refinement near a good anchor: by Eq. (21), the sufficient detection guarantee at $h = 0.005$ uses about 3.18×10^6 shots per endpoint, roughly twenty-four times the largest per-run optimization budget in this study, and about forty-eight times it for the endpoint pair.

Proposition 5 (Acceptance probability below the confidence allowance). *Fix the history $\mathcal{H}$ after selecting a trial and before drawing its acceptance samples. Let $h = a - \eta q$ be its true acceptance margin. Suppose all observations within and across the two fresh endpoint streams are conditionally independent given $\mathcal{H}$, lie in $[-1, 0]$, and have the corresponding true objective means. For the Hoeffding rule in Eq. (18),*

$$\mathbb{P}(\text{accept this trial} \mid \mathcal{H}) \leq \min \left\{ 1, \sum_{\ell=1}^L \exp \left[-s_{\ell} (2b_{\ell} - h)_+^2 \right] \right\}, \quad (22)$$

where $(t)_+ = \max\{t, 0\}$. The bound remains valid when early rejection or an insufficient budget prevents some prescribed looks.

Proof. At look ℓ , $\hat{a}_{\ell} - a$ is a centered sum of $2s_{\ell}$ independent terms, each with range length $1/s_{\ell}$. The sum of squared range lengths is $2/s_{\ell}$. The one-sided form of Hoeffding’s inequality [18] therefore gives $\mathbb{P}(\hat{a}_{\ell} - a \geq t \mid \mathcal{H}) \leq \exp(-s_{\ell} t^2)$ for $t \geq 0$. Acceptance requires $\hat{a}_{\ell} - a \geq 2b_{\ell} - h$. Apply this inequality when the threshold is positive and use the trivial bound one otherwise. Union bound over the prescribed looks, extending the fresh streams to unexecuted looks if necessary. Removing looks cannot add acceptance events, and independence between cumulative looks is unnecessary. $\square$

For the canonical four-look schedule, every trial with $h \leq 0.005$ has conditional acceptance probability at most 3.48×10^{-6} , even if all four looks are affordable. This is an upper bound for the specified policy; it does not constrain other estimators or acceptance rules. Together with Corollary 4, it distinguishes valid decisions from a rule’s ability to detect small improvements.

The canonical policy allows $T = 64$ trials with $\eta = 0.1$ and $\alpha = 0.05$. It examines each endpoint after 512, 2048, 8192, and 32768 cumulative shots. New batches contribute only the difference from the preceding look. If $\hat{a}_{\ell} + 2b_{\ell} < \eta q$, the trial is rejected for failing the required decrease on the confidence event. Otherwise additional shots are requested until a decision or a budget limit. Unresolved trials leave the center unchanged. Confidence is per optimization run; 95% does not mean simultaneous validity across the complete collection of benchmark runs. Methods and budgets reuse seed labels to support paired comparisons; their outcomes are not assumed independent.

The canonical initial radius is 0.4. Acceptance multiplies it by 1.5 up to 0.8; rejection or an unresolved decision multiplies it by 0.5 down to 0.025. A predicted decrease below 10^{-12} terminates the run. A fixed-radius ablation retains 0.4 throughout. The fixed-shots ablation uses only the final 32768-shot look, with $L = 1$ in the confidence bound. This is a prespecified sampling policy, not a hindsight-matched optimal fixed allocation. Its single look costs 65536 shots for the endpoint pair, so no run affords a second trial at either canonical budget: at 32768 shots all 144 fixed-shots runs fit one model and stop before any acceptance measurement, and at 131072 shots all 144 complete exactly one acceptance test, every one of which rejects. This ablation therefore accepts no step anywhere in the study. Such costs and unresolved decisions are retained in the records.

Algorithm: one local-model optimization run. The inputs are the selected anchor x , shape B , shot cap, and the settings above. The output is the last accepted incumbent and all measurement and decision records.

1. Initialize $\Delta = 0.4$, spent shots to zero, and x as the zero-shot incumbent. Use the four prescribed looks, or only the final look for fixed shots, and set L accordingly. Repeat Steps 2–6 for at most $T = 64$ trials.
2. Select $d + 1$ design points by QR, or use the raw random points for no repair. Report a numerical failure if the design is rank deficient. If its complete 256-shot-per-point batch exceeds the remaining budget, return x . Otherwise sample and charge that batch, fit m by weighted QR, and record its condition number.
3. Put $q = \|\hat{c}_{1:d}\|_2$. If $q < 10^{-12}$, terminate with a flat-model status. Otherwise set $x^+ = \text{wrap}(x - \Delta B \hat{c}_{1:d}/q)$. Initialize the decision to unresolved, both endpoint sample sums to zero, and the previous cumulative sample count to zero.
4. For each prescribed s_ℓ , request only the new samples needed at both endpoints. If the complete pair exceeds the remaining budget, end this acceptance loop without a partial batch. Otherwise charge the batch and compute $\hat{a}_\ell$ and b_ℓ from Eq. (17). Accept and end the loop if $\hat{a}_\ell - 2b_\ell \geq \eta q$; reject and end it if $\hat{a}_\ell + 2b_\ell < \eta q$.
5. Record the decision, including unresolved outcomes. On acceptance set $x \leftarrow x^+$ and record its cumulative shot count. With radius adaptation, expand by 1.5 up to 0.8 after acceptance, or contract by 0.5 down to 0.025 otherwise. Fixed radius leaves Δ unchanged.
6. Terminate if the remaining budget cannot cover another model batch and the first acceptance look at both endpoints; otherwise continue. Every ordinary stopping condition returns the incumbent, including the original anchor if no step was accepted.

Rejected and unresolved trials therefore retain the current incumbent. A rank failure, nonfinite structural coupling, or failed marginal check instead raises a numerical error; the implementation does not silently substitute another graph score. An unaffordable later acceptance look need not terminate the outer loop if a new model and first look remain affordable.

4.7 Canonical instances, reference bank, and comparators

The canonical configuration fixes the master seed to 20260918. NetworkX generators [16] produce independent-edge $G(n, 0.4)$ graphs [13], random 3-regular graphs using the Steger–Wormald generator [33], Barabási–Albert attachment graphs with two new edges per vertex [1], and Watts–Strogatz rewiring graphs with initial degree four and rewiring probability 0.35 [37]. Disconnected draws are rejected. The reference bank contains two graphs from each family at each of $n = 8, 10$, giving 16 graphs. The test set contains three per family at each of $n = 10, 12$, giving 24 graphs. Every accepted graph is nonisomorphic to all earlier accepted graphs in either split, including across family labels. These filters modify the base sampling distributions; exact uniformity over the retained regular graphs is not assumed. Saved edge lists are the definitive instances. The study examines $p = 1, 2$ separately, with a separate parameter bank at each depth.

For each graph and depth, six BFGS runs [26] on the exact expectation supply a numerical reference: one deterministic angle schedule and five uniformly randomized starts in the symmetry cell. The SciPy implementation [36] estimates gradients by two-point finite differences because no Jacobian is supplied. BFGS uses at most 180 iterations and gradient tolerance 10^{-7} . For every

Table 4: Prespecified comparators and ablations. All local-model variants share the QR solver and finite-horizon acceptance rule except where a sampling or radius change is stated.

Variant	Controlled choice
Fixed donor	First bank anchor, isotropic shape
Random start	Uniform random angle vector in the symmetry cell, isotropic shape
Descriptor, isotropic	Descriptor-selected anchor, $B = I$
Descriptor, shaped	Same anchor, reference scatter shape
GW, isotropic	GW-selected anchor, $B = I$
GW, shaped	Same anchor, reference scatter shape
Local TV	Rooted-neighborhood total-variation score, $B = I$
Shuffled GW	Permuted GW scores before anchor and shape
Displaced prior	Descriptor anchor shifted by $(1.2\mathbf{1}_p, 0.35\mathbf{1}_p)$ and wrapped, reference scatter shape
No point repair	Descriptor shape, random interpolation points
Fixed radius	Descriptor shape, radius fixed to 0.4
Fixed shots	Descriptor shape, one final acceptance look
COBYLA	Descriptor anchor, 1024 shots per estimate
SPSA	Descriptor anchor, 1024 shots per estimate

start we retain the returned angle vector, its objective value, the convergence flag, the termination message, and the exact evaluation count, together with one construction time per graph and depth. The starting vectors themselves are not stored; they are fixed by the recorded configuration and seed. The lowest recorded value is f_G^{ref} ; it is not asserted to be globally optimal. Test-graph references are diagnostic only and are not available to the optimizers or graph-prior construction. The signed reference gap $f_G(\theta) - f_G^{\text{ref}}$ may be negative.

Fourteen methods are specified before measurement trials. Table 4 gives their controls. All isotropic/shape pairs use the same selected anchor. The shape, no-repair, fixed-radius, and fixed-shots variants share the descriptor-selected anchor unless explicitly identified as GW variants. For each graph, depth, and method, three shot seeds are run at optimization budgets 32768 and 131072. All accepted and unsuccessful trials count toward the budget. The public configuration determines the complete study rather than a selection of successful runs.

COBYLA uses Powell’s derivative-free method [29] through the PRIMA implementation [39] included in SciPy 1.17.1 [36], with initial radius 0.4 and tolerance 0.002, subject to the same shot cap. Its `maxiter` option bounds objective evaluations. SPSA uses the simultaneous-perturbation gradient estimator [32] with independent Rademacher perturbations. We set the gains to $a_k = 0.15/(k+10)^{0.602}$ and $c_k = 0.12/k^{0.101}$ and use two perturbed estimates plus one additional incumbent estimate per update. Both retain the point with the smallest observed incumbent estimate. Their true-objective trajectories need not be monotone. These baseline settings are fixed rather than tuned on the test results; a different competent tuning may change their ranking.

4.8 Follow-up across two reference banks

The follow-up is a separate experiment designed after examining the canonical study. Its full configuration is `code/configs/followup.json`. Two panels use seeds 2026091901 and 2026091902. Each contains eight bank graphs (the four families above at $n = 8, 10$), four validation graphs (one per family at $n = 12$), and eight test graphs (the four families at $n = 12, 14$). All 40 accepted graphs are mutually nonisomorphic. Cross-study inspection finds one 12-vertex regular follow-

up test graph isomorphic to a canonical test graph; the other 15 test graphs are nonisomorphic to every canonical instance. This overlap is retained and disclosed rather than removed after observing outcomes. One eight-vertex regular bank graph also matches a canonical bank graph; the other 15 bank graphs and all eight validation graphs are nonisomorphic to every canonical instance. No follow-up bank or validation graph matches a canonical test graph. Every target uses only its own panel’s bank. The two panels are separately seeded replications with an isomorphism exclusion across panels, rather than unrestricted independent graph draws.

The circuit depths are $p = 2, 3$. References use six BFGS starts, tolerance 10^{-7} , and at most 120 iterations per start. As in the canonical study, these are best-found references, with every exact objective call retained; no global-optimality claim is made. The seven test policies are descriptor initialization alone, a fixed graph-independent schedule, matched isotropic Hoeffding refinement, isotropic and shaped empirical-Bernstein refinement, SPSA, and COBYLA. The fixed schedule is $\gamma = \text{linspace}(0.35, 0.8, p)$ and $\beta = \text{linspace}(0.35, 0.15, p)$. Initialization-only and fixed-schedule policies make no optimization calls. All other policies start from the same descriptor-selected anchor. Only the shaped Bernstein variant uses $B \neq I$.

Each run has an optimization cap of 65536 shots and receives 8192 independent final-validation shots. At each depth, the validation phase compares four combinations of model shots $S \in \{256, 1024\}$ and initial radius $\Delta \in \{0.1, 0.3\}$, using both Bernstein shapes on all eight validation graphs. This gives 128 validation runs. The selection criterion is the mean finite-shot final-validation objective over graphs and the two shapes; exact expectations and test outcomes are not used. Candidate order breaks ties. All validation runs exhaust the optimization budget without accepting a trial, so each returns its initial anchor; because the final-validation seed is derived from the run seed, which does not depend on the candidate, the four candidates draw the same 8192 shots at the same angle and the four scores are bitwise identical at each depth. Thus $S = 256$ and $\Delta = 0.1$ are selected by the tie-break at both depths, not by an observed quality improvement. The selected settings and validation records are hashed before the 448 test runs: seven methods, 16 graphs, two depths, and two shot seeds. Paired methods reuse seed labels; their results are not independent within a graph. The shared local-model settings enable a controlled comparison of acceptance rules at $B = I$. No shaped Hoeffding policy is run, so the two-by-two design of isotropic and shaped regions crossed with Hoeffding and empirical-Bernstein acceptance is missing one cell. The acceptance rule is therefore varied only at $B = I$ and the shape only under empirical-Bernstein acceptance; the shape-by-rule interaction is not identified, and a shaped Hoeffding region remains untested. SPSA retains its gains, and both SPSA and COBYLA use 1024 shots per estimate. COBYLA retains tolerance 0.002 but uses the common selected initial radius 0.1, instead of the canonical value 0.4. Neither baseline receives a separate tuning search.

The follow-up retains $\eta = 0.1$ and $\alpha = 0.05$, uses $T = 16$ trials, and prescribes cumulative looks (512, 2048, 8192, 16384) per endpoint. The radius contracts by 0.5 or expands by 1.5 with the same bounds $[0.025, 0.8]$. Fresh endpoint streams are sampled after the model proposes a trial. For an endpoint with $s \geq 2$ observations, let $\hat{v}_s$ be the unbiased sample variance of individual outcomes. The established empirical Bernstein inequality of Maurer and Pontil [23, Theorem 4] gives the radius

$$r_s = \sqrt{\frac{2\hat{v}_s \log(8TL/\alpha)}{s}} + \frac{7 \log(8TL/\alpha)}{3(s-1)}. \quad (23)$$

Apply their one-sided inequality to $Y + 1 \in [0, 1]$ and its reflection, assigning failure probability $\alpha/(4TL)$ to each endpoint and tail. Conditioning on the history before each trial and taking a union bound over all prescribed trials and looks gives simultaneous interval validity with probability at least $1 - \alpha$ per run. The decrease interval has radius $r_{x,s} + r_{+,s}$: acceptance requires $\hat{a}_s - r_{x,s} - r_{+,s} \geq$

ηq , and rejection requires $\hat{a}_s + r_{x,s} + r_{+,s} < \eta q$. Other outcomes remain unresolved. This is a direct finite-horizon application of an existing inequality, not a new confidence bound. It preserves the true-decrease conclusion of Theorem 3 under the same sampling assumptions.

Cumulative sample variances combine each batch’s within-batch sum of squares with the between-batch correction. Averaging the recorded variances of batch means would not give the required $\hat{v}_s$. The comparator models uncertainty in acceptance measurements; it does not fit a spatial variance model or reproduce ASTRO-DF [15]. The targets $f_G^{\text{ref}} + \epsilon$ for $\epsilon \in \{0.002, 0.005, 0.01, 0.02\}$ are fixed in the follow-up configuration before its test phase. Their hitting costs use recorded incumbents and exact diagnostic expectations; they never stop or steer the optimizer. Table 2 summarizes the complete test sample. The two-bank structure and small graph sample restrict generalization; per-panel summaries accompany pooled graph-level intervals. Each panel draws its bank, validation split, test split and random streams together, so a difference between panels is not attributable to the reference bank alone.

4.9 Resource accounting, validation, and reproducibility

An objective-estimation call is one newly sampled circuit estimate at one angle vector. A shot is one repeated circuit execution. A job is one logical batch submitted by the algorithm; a model batch contains all interpolation points, and an acceptance batch contains the two endpoints. Additional samples at an old point constitute a new estimation call. The oracle rejects a batch atomically if it would exceed the remaining shot budget. It records every call, batch, shot count, angle, sample mean, estimated variance of the mean, and elapsed simulation time. Budget caps are limits, not an assertion that every run spends exactly the same number of shots.

In the canonical study, every final vector receives 16384 independent validation shots, recorded as one additional call and job outside the optimization budget. Total operational shot cost is the optimization count plus this validation cost. Validation measurements never select the returned vector. The primary comparison uses the exact expected objective at the returned vector; validation means quantify the finite-shot evaluation available to an experimental user. The canonical target is $f_G(\theta) \leq f_G^{\text{ref}} + 0.02$. Hitting cost is the first recorded incumbent’s cumulative optimization shot count satisfying this condition. Runs that never hit the target remain failures, and their costs are retained. The budget-capped target cost assigns the full optimization cap to failures; for successes it uses the first hitting cost. This summary differs from actual incurred shots when a method stops early.

Repeated seeds for one graph do not create additional independent graph instances. The empirical summaries retain the graph as the comparison unit, report depth and budget separately, and distinguish successful hitting costs from success fractions. Classical reference construction and structural preprocessing are reported separately from online simulation time. Simulator timings are not estimates of quantum hardware execution or communication latency. Intervals use the percentile bootstrap [9]: a custom NumPy routine averages shot repetitions within each graph, resamples graph means 10000 times with seed 5713, and takes the 2.5th and 97.5th percentiles of the resampled means. Paired contrasts resample within-graph differences. The follow-up resamples within each of its two fixed reference banks; these intervals describe variability across the observed graphs, not uncertainty over a population of banks. They are descriptive, unadjusted intervals, without bias correction or acceleration.

The implementation uses NumPy 2.4.3 for array computation and sampling [17], SciPy 1.17.1 for optimization and linear algebra [36], NetworkX 3.6.1 for graph construction and comparison [16], and Matplotlib 3.10.8 for figures [20]. The environment versions and source hashes saved with the run data identify the implementation used for the reported computations.

The accompanying tests compare the statevector against independently constructed dense Pauli-matrix exponentials, verify the analytic single-edge objective, enumerate cuts independently, and compare weighted QR fits with an SVD-based least-squares solution. Additional checks cover exact graph isomorphism in neighborhood comparison, angle symmetries, coupling marginals and direct four-index distortion, shape determinant and conditioning, budget accounting, and deterministic repeatability. These checks establish specified implementation properties; they do not establish empirical superiority or general optimizer convergence.

4.10 Scientific corrections and experiment provenance

This manuscript replaces the earlier Gaussian-predictor formulation and its empirical aggregates with the independently implemented study described here. Historical aggregate evaluation, timing, calibration, and noise results are not retained because the recovered materials did not include the underlying experiment records needed to reproduce them. The earlier best-of- K derivation also contained an algebraic error. Let $D_k = \|X_k - \mu\|$ for identically distributed draws $X_1, \dots, X_K$ with mean μ and covariance Σ , so that $\mathbb{E}D_k^2 = \text{tr} \Sigma$ for every k . Then $\sqrt{K^{-1} \sum_{k=1}^K \mathbb{E}D_k^2}$ equals $\sqrt{\text{tr} \Sigma}$, not $\sqrt{\text{tr} \Sigma / K}$: averaging K identically distributed squared distances leaves the root-mean-square distance unchanged and supplies no $K^{-1/2}$ factor. No such rate is claimed here. The present method still selects a single donor from the reference bank by maximizing a prior weight, but no part of the analysis credits that selection with an improvement that grows with the number of bank references. The new empirical record is not a reconstruction of the historical figures.

Data availability

The graph instances, best-found reference labels, complete per-run records (calls, batches, shot ledgers, trial decisions, incumbents, and final validation), transport-audit records, follow-up records, aggregate summaries, figure provenance, and the recorded environment are included in the accompanying code and data folder, with an input manifest linking the records to generated outputs. The canonical, follow-up, and transport record files are the authoritative sources for every number reported here; the tables and figures in this manuscript are regenerated from them. Exact simulator diagnostics are stored alongside the shot-based measurements and are labelled as diagnostic.

Code availability

The simulator, transfer and search implementations, experiment runners, analysis and figure generators, the independent record verifier, and the artifact checker are included in the accompanying code folder. The verifier reconstructs saved model fits, acceptance decisions, shot ledgers, and hitting times from the saved records; the test suite checks the statevector against independently constructed dense Pauli exponentials, the analytic single-edge objective, exact angle symmetries, transport marginals and a direct four-index distortion, QR against SVD least squares, and budget accounting. These materials are supplied for inspection and reproduction. They are not offered under an open-source or other reuse license; the accompanying notice states the terms.

Author contributions

M.H. designed and implemented the study, performed the experiments and analysis, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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The computations use NumPy, SciPy, NetworkX, and Matplotlib.

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